EXP17: EYE DIAGRAM ANALYSIS OF OPTICAL COMMUNICATION LINK USING OPTISYSTEM

OBJECTIVES:

1. Design and establish a fiber optic communication link using OptiSystem software.

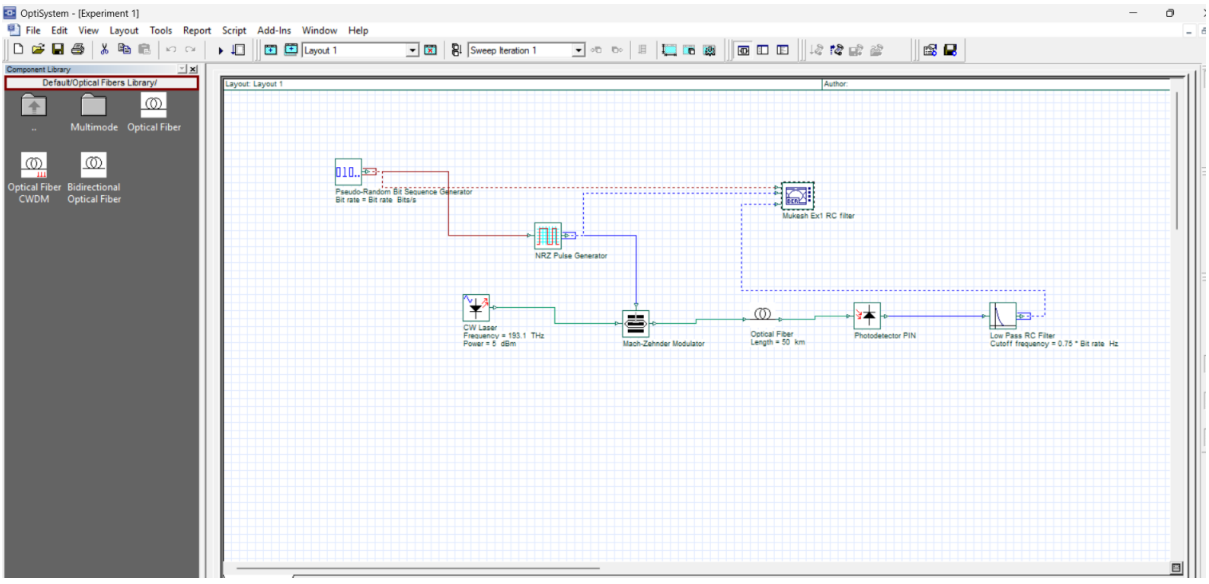
2. Measure and evaluate the Bit Error Rate (BER) of the system for different fiber lengths.

3. Study the effect of fiber length on BER, Q-factor, a enuation, and dispersion.

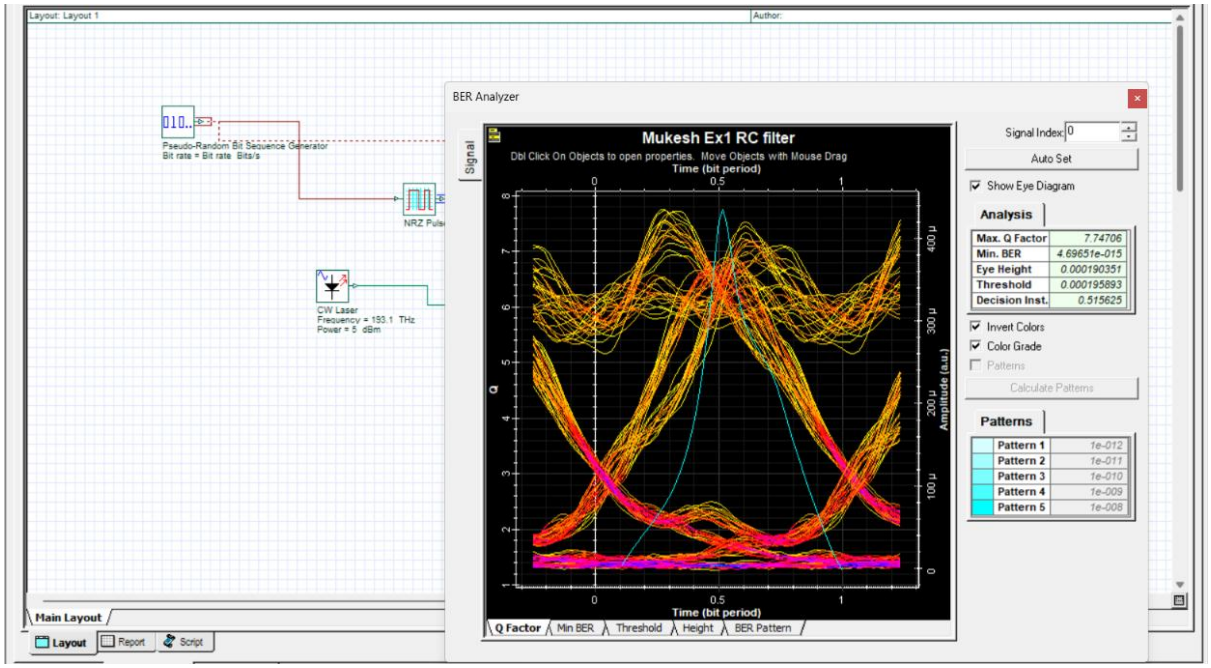
SOFTWARE REQUIRED

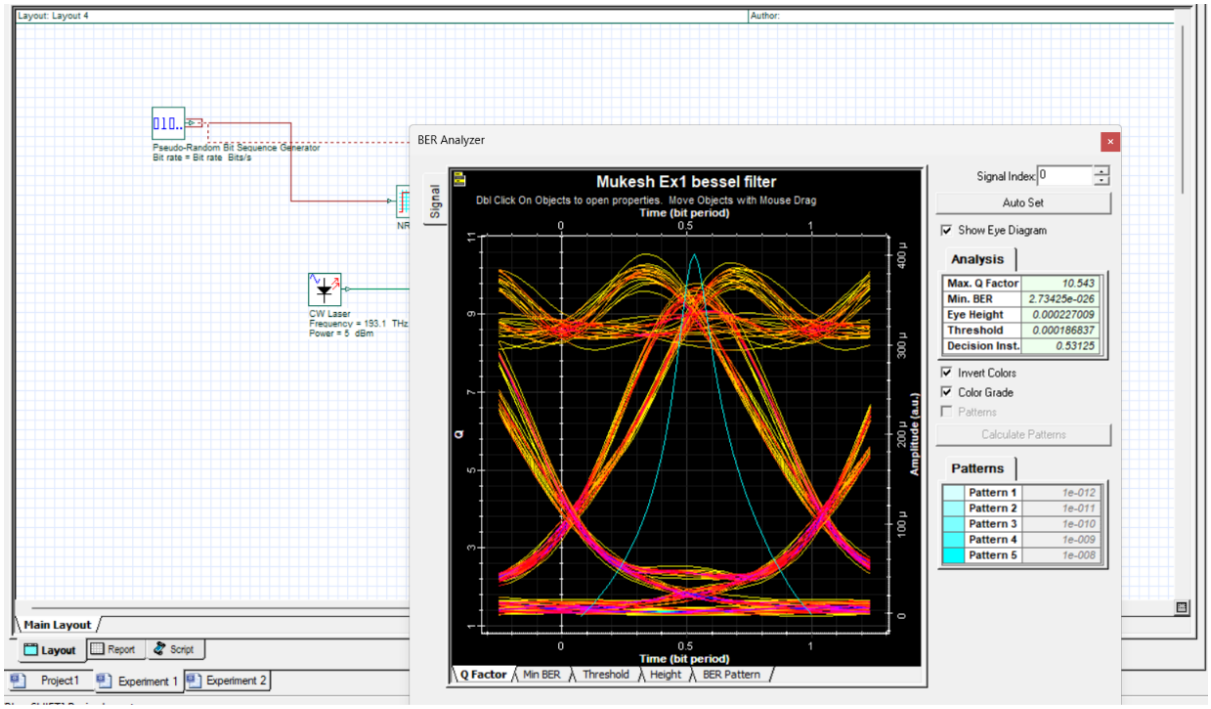
• Optisystem software

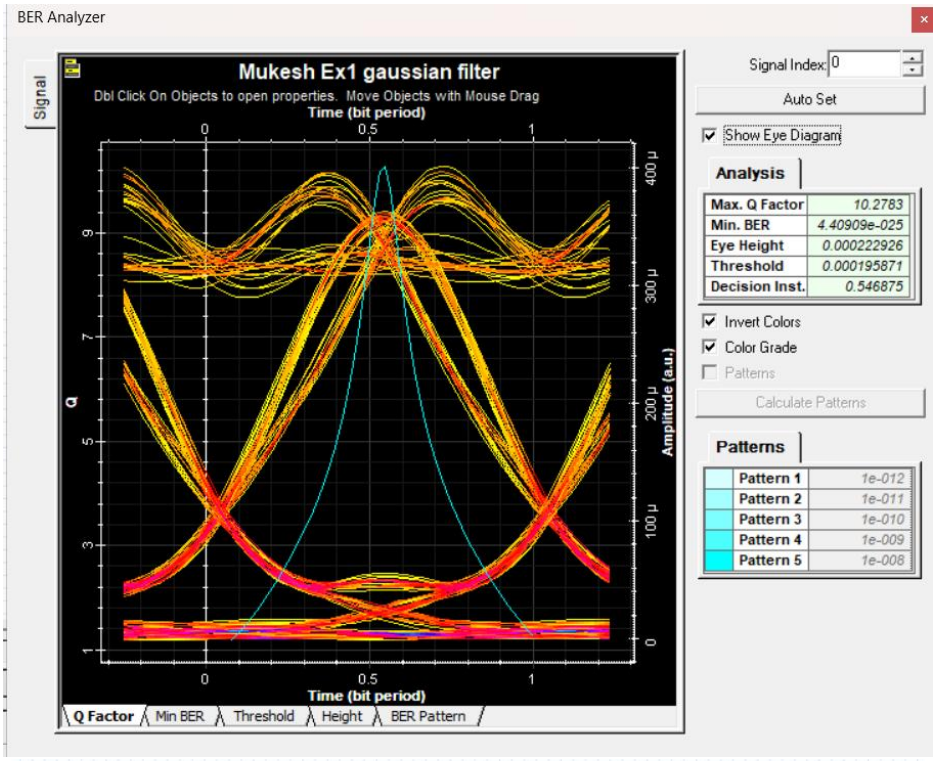
BLOCK DIAGRAM:

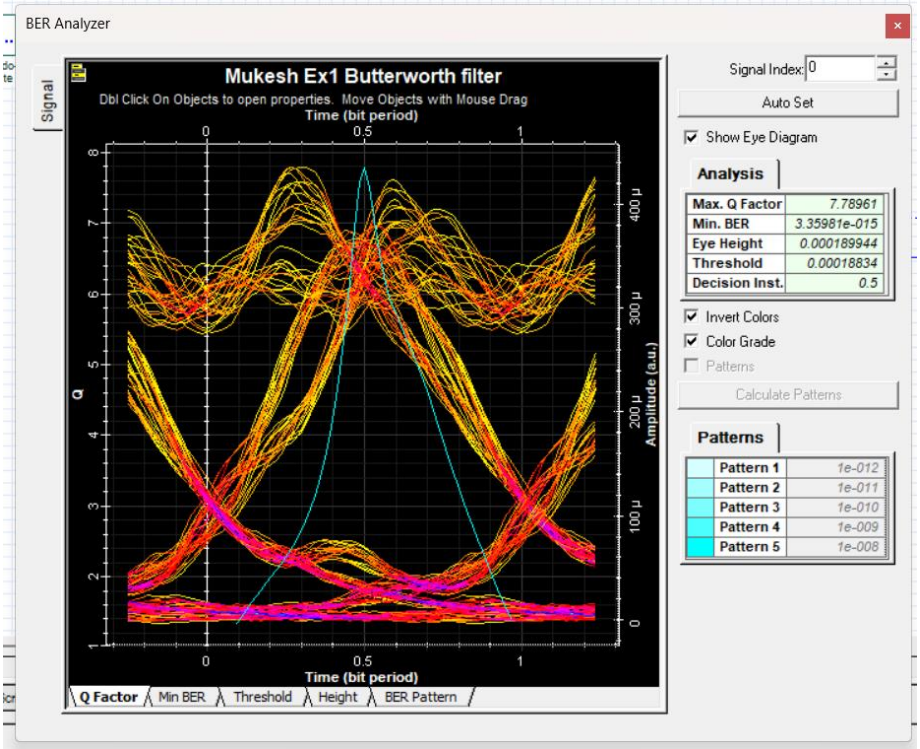


Outputs:


Layout: Layout 1
Author:
Pseudo-Random Bit Sequence Generator
Bit rate = Bit rate Bits/s
NRZ Pulse
CW Laser
Frequency = 193.1 THz
Power = 5 dBm
Main Layout
Layout
Report
Script
BER Analyzer
Mukesh Ex1 RC filter
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Signal
Q
Amplitude (a.u.)
Signal Index 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 7.74706
Min. BER 4.69651e-015
Eye Height 0.000190351
Threshold 0.000195893
Decision Inst. 0.515625
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008
Q Factor
Min BER
Threshold
Height
BER Pattern


Layout: Layout 4
Author:
Pseudo-Random Bit Sequence Generator
Bit rate = Bit rate Bits/s
CW Laser
Frequency = 193.1 THz
Power = 5 dBm
Main Layout
Layout
Report
Script
Project1
Experiment 1
Experiment 2
BER Analyzer
Mukesh Ex1 bessel filter
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Signal
Q
Amplitude (a.u.)
Signal Index 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 10.543
Min. BER 2.73425e-026
Eye Height 0.000227009
Threshold 0.000186837
Decision Inst. 0.53125
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008
Q Factor
Min BER
Threshold
Height
BER Pattern


BER Analyzer
Signal
Mukesh Ex1 gaussian filter
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Q
Amplitude (a.u.)
Signal Index 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 10.2783
Min. BER 4.40909e-025
Eye Height 0.000222926
Threshold 0.000195871
Decision Inst. 0.546875
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008
Q Factor
Min BER
Threshold
Height
BER Pattern


BER Analyzer
Signal
Mukesh Ex1 Butterworth filter
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Q
Amplitude (a.u.)
Signal Index 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 7.78961
Min. BER 3.35981e-015
Eye Height 0.000189944
Threshold 0.00018834
Decision Inst. 0.5
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008
Q Factor
Min BER
Threshold
Height
BER Pattern

OUTPUT PARAMETERS:

S.No	Filter name	Max Q factor	Min.BER	EYE Height	Thereshold
1	RC	8.23399	8.94569e-017	0.000196077	0.000206332
2	Butterworth	7.78961	3.35981e-015	0.000189944	0.00018834
3	Bessel	11.3849	2.47064e-030	0.000233284	0.00020034
4	Gaussian	10.2833	4.18317e-025	0.000223866	0.000194259

CONCLUSION:

The study shows how fiber length affects BER and signal quality in an optical communication system.

• 25 km fiber: highest Q-factor (15.26) and lowest BER (7.06×10^{-53}), with a wide open eye diagram.

• 50 km fiber: reduced Q-factor (8.02) and BER of 5.26×10^{-16} due to a enuation; the link is still reliable ($Q > 6$, $BER < 10^{-9}$).

• 75 km fiber: Q-factor falls to 4.89 and BER rises to 5.00×10^{-7} as a enuation and dispersion close the eye; the link is no longer reliable.

• 100 km, 125 km and 150 km fiber: complete link failure (Q-factor = 0, BER = 1, eye height = 0). The maximum reliable length is therefore between 50 km and 75 km.

Reflection:

I learned to design and simulate an optical communication link using OptiSystem. I understood how fiber length affects the eye diagram, Q-factor and BER. This experiment helped me analyze signal quality and understand the performance of optical communication systems.

LABORATORY PERFORMANCE ASSESSMENT RUBRIC

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
System Design & Fiber Link Setup	Correct selection and connection of OptiSystem components for the optical fiber link (PRBS Generator, NRZ Pulse Generator, CW Laser, Optical Fiber, PIN Photodetector, Low Pass Filter, 3R Regenerator, BER Analyzer).	All transmitter, fiber, receiver, and analyzer components are correctly selected, connected in sequence, and configured without errors.	All components are present with minor connection or layout errors, but the link remains functional.	Component selection is partially complete; minor connection or configuration errors impede signal flow.	Incorrect component selection or improper setup leading to complete simulation failure.	20%
Parameter Configuration & Execution	Accurate setup of input parameters: Optical Power (5 dBm / 10 dBm), Frequency (193.1 THz), Wavelength (1550 nm), Bit Rate (10 Gbps), and Fiber Length (25 to 150 km).	All input parameters and fiber length test cases are accurately set and simulated.	Test case parameters are mostly accurate with minor setting discrepancies.	Basic parameters are configured, but multiple fiber length cases are missing.	Incorrect parameter settings or failure to run the required fiber length cases.	20%
Measurement & Eye Diagram Observation	Accurate extraction and recording of Max. Q-Factor, Min. BER, Eye Height, Threshold, and Decision Instant from the BER Analyzer for each fiber length.	Precise data extraction for all fiber lengths, neatly tabulated, with clear eye diagrams captured for each test case.	Data is collected with minor omissions in Q-factor, BER, or eye parameters.	Partial data collection with incomplete tables or missing eye diagrams.	Inaccurate, incomplete, or missing data and measurements.	20%
BER & Signal Degradation Analysis	Detailed evaluation of the effect of fiber length on BER, Q-factor, eye opening, attenuation, and dispersion.	Comprehensive analysis correctly relating fiber length to attenuation, dispersion, eye closure, Q-factor decline, and BER increase.	Clear analysis identifying basic trends in Q-factor, BER, and eye height versus fiber length.	Basic attempt at result analysis without explaining physical mechanisms such as attenuation and dispersion.	Erroneous or missing analysis of simulation results and eye diagrams.	20%
Report & Conclusion	Identification of the maximum reliable transmission distance, comparison across short, medium, long, and very long links, and structural clarity.	Report features clear eye diagrams, complete output tables, and insightful conclusions identifying the reliable link length limit.	Satisfactory report with relevant eye diagrams and sound concluding summary.	Incomplete report with missing diagrams, tables, or weak conclusions.	Disorganized or incomplete report lacking required diagrams and valid conclusions.	20%